

Geospatial proposal assessment

USD describes a 3D model's shape and size. Geospatial information says where it belongs on Earth, so models, surveys and measurements can work together.

The full follow-up now starts from placement authoring: a readable heading/pitch/roll input, with explicit physical meaning and sample evaluation. The focused requirements baseline remains separate.

All 105 prior case comparisons remain within their existing acceptance limits. New orientation and composition controls pass. Normal sampling remains unresolved; the preferred HPR model still needs group adoption.

A geographic location for a 3D model

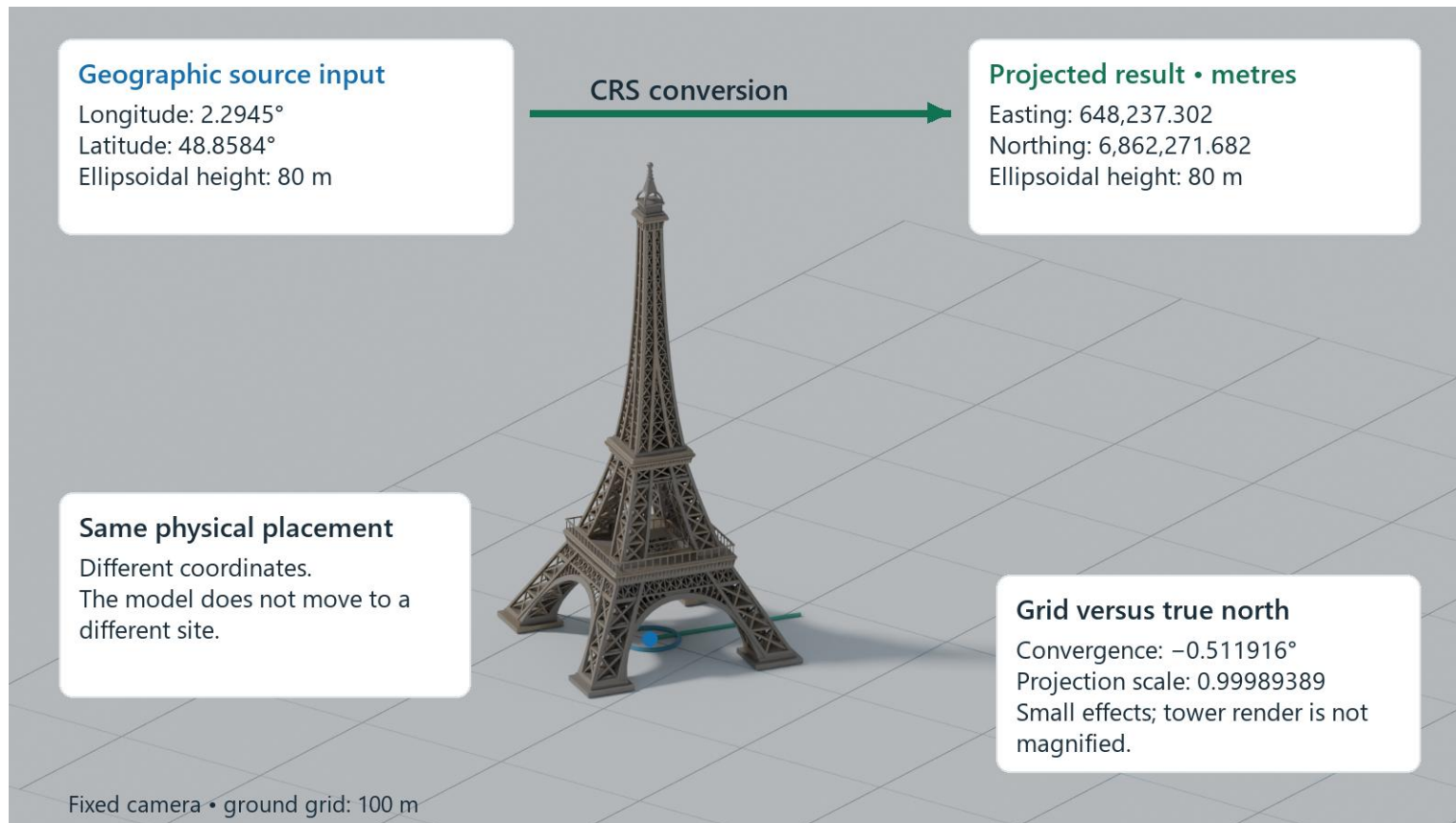
Can a reader distinguish the model's shape from its location on Earth?

Source information	Plain meaning
Model geometry	Shape and local distances, with declared model units and up direction. In USD: points, metersPerUnit and upAxis.
Coordinate reference	A CRS binding refers to WKT, the standard text description of the location's axes, units and height meaning.
Geographic origin	crs:position locates the model origin. The illustrative example uses longitude 2.2945 degrees, latitude 48.8584 degrees and ellipsoidal height 80 m.
Physical orientation	crs:orientation stores heading, pitch and roll in degrees: which way the model faces and tilts at its origin.

A reader has enough source information to place the model, without interpreting longitude or latitude as a distance or adding computed results to the source schema.

One physical place, different coordinates

Does changing coordinate systems preserve physical placement?



Observed

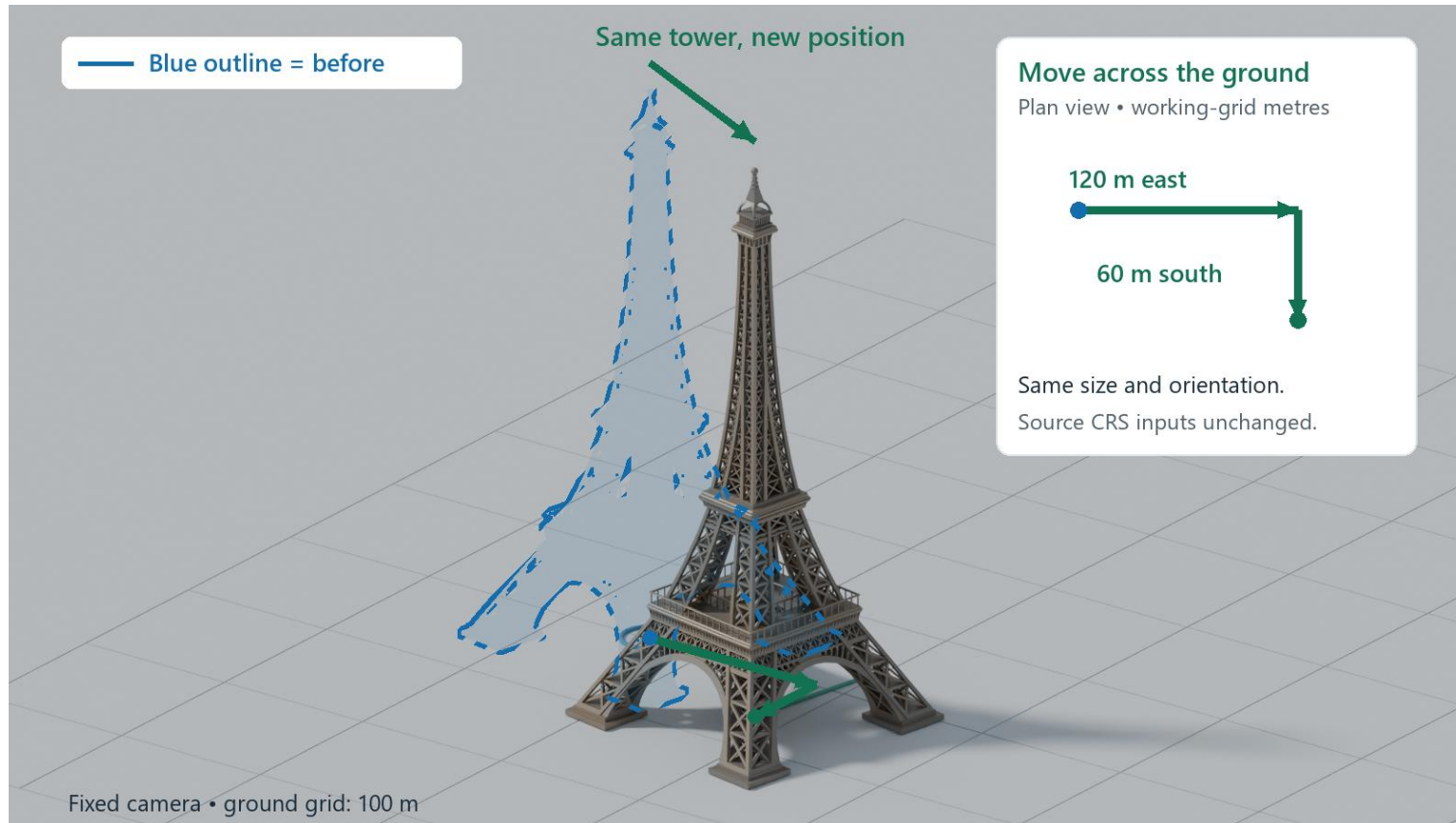
The illustration converts longitude, latitude and height into map coordinates. The physical site stays the same; map north and local scale can differ.

What it establishes

The proposed rules distinguish a new coordinate representation from a physical move. An image explains this distinction; numerical controls must test it.

Ordinary model adjustments after placement

In which coordinates do an author's scale, rotation and translation operate?



Observed

The illustration applies scale 1.15, rotation 20 degrees, then moves 120 m east and 60 m south. A child-transform control separates two interpretations by 4.222 m.

What it establishes

Order changes the result. The control tests the agreed anchor/child distinction while detailed chart and instance conventions remain candidates.

The same rules in two placement readers

Can separate implementations derive the same coordinates from the proposed inputs and rules?

France: CC49 to Lambert-93, metres

Resolved component	Python reader	C++ reader
Map east coordinate	661,101.109417	661,101.109417
Map north coordinate	6,857,834.256536	6,857,834.256536
Height	37.298000	37.298000

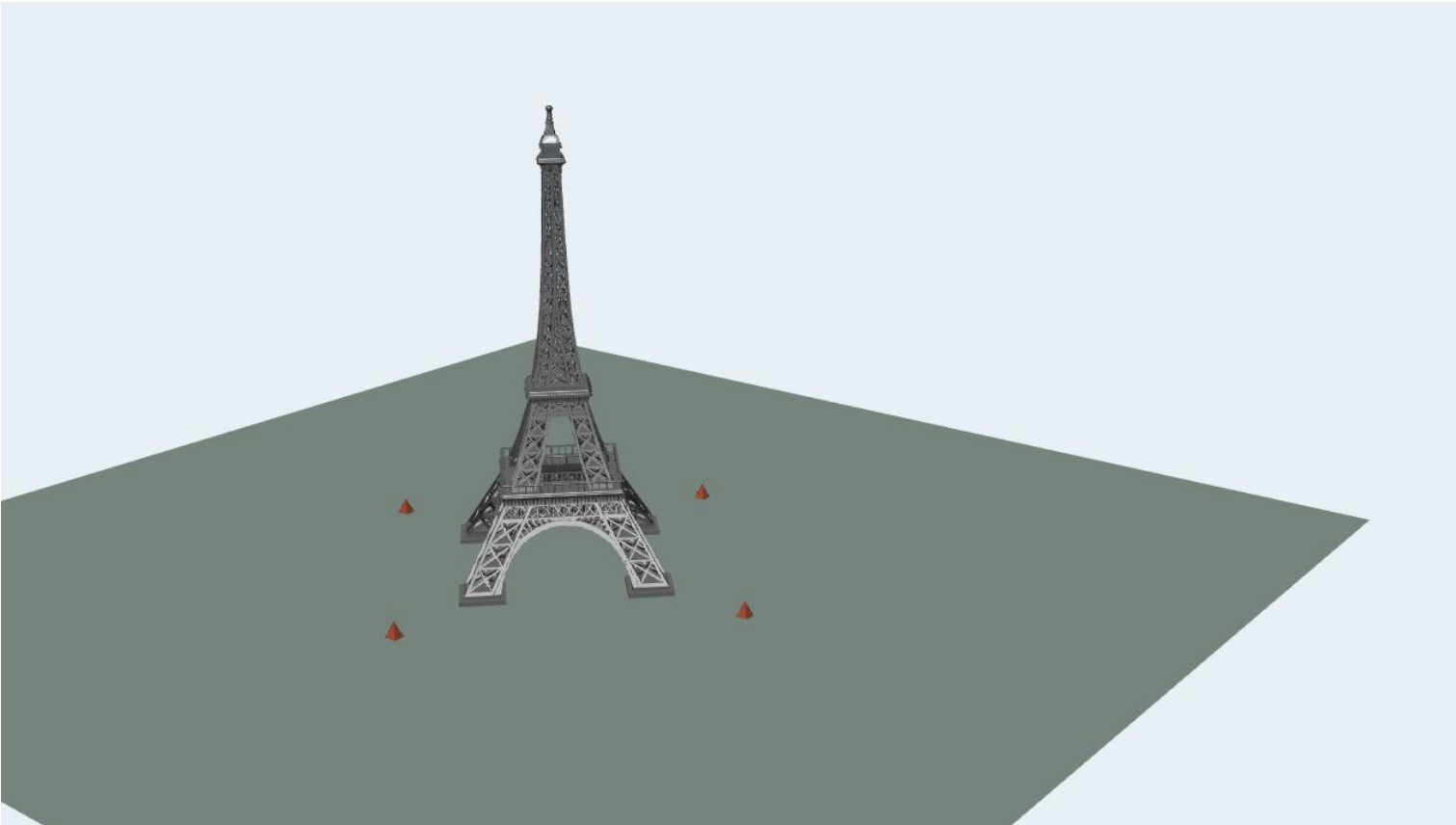
Colorado: State Plane to site grid, US survey feet

Resolved component	Python reader	C++ reader
Map east coordinate	3,108,871.450101	3,108,871.450101
Map north coordinate	1,206,317.469403	1,206,317.469403
Height	5,484.677949	5,484.677949

Both readers use PROJ and share dataset decoding.

Ordinary USD geometry for rendering

Can a renderer use the result without implementing geospatial evaluation?



Observed

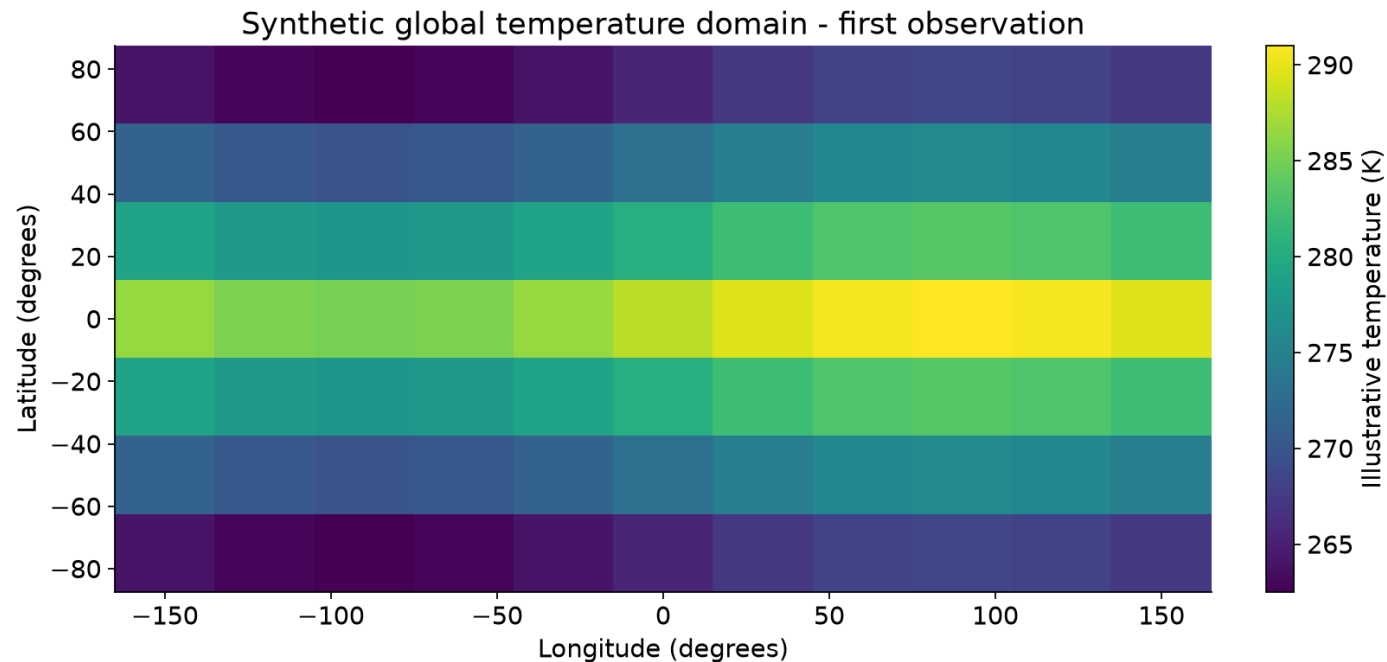
Native Hydra/Storm renders the ordinary USD bake. A double-precision origin keeps local points small; fresh readers recover the placement.

What it establishes

Unaware viewers receive resolved ordinary geometry. Fresh checks reject duplicate meshes and retain an unrelated applied schema.

Measurements stay paired with their locations

Can the scene make a native measurement dataset usable without rewriting its values or guessing what its coordinates mean?



The synthetic global temperature case returns both coordinate forms for 154 samples and preserves the native values and two observation times.

The values are synthetic. Named-dimension associations are independently tested; agreeing readers alone cannot verify shared decoding.

Fresh independent audit

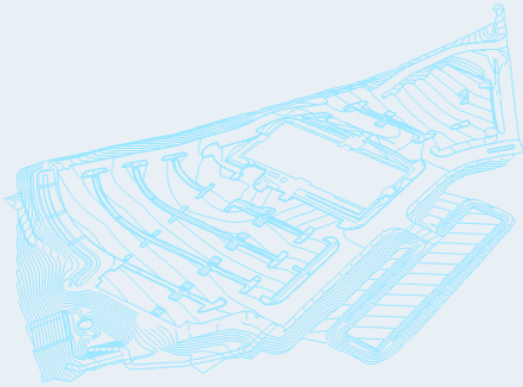
Six prior defects pass fresh counterexamples; one remains unresolved.

Affected behavior	Fresh audit result
Surface lighting directions	Remaining: normal sampling has no justified domain/error bound; projected difference about 1.24 degrees.
Repeated models	Corrected: an unbound library prototype receives its instancer placement in both readers.
Baked geometry	Corrected: two source meshes yield exactly two baked meshes; extras are rejected.
Measurement locations	Corrected: all six measurements retain the coordinates declared by dimension names.
Other scene information	Corrected: the bake retains MotionAPI and its velocity-scale value.
Error reporting	Corrected: reports include component errors, magnitudes and angular errors.
Invalid input detection	Corrected: both readers reject an invalid child placement field.

Normal sampling still needs justified domain/error evidence. Do not turn its shortcut into a proposal rule.

Appendix: A construction site's own survey grid

Can a project's local survey grid participate as a defined CRS?



Observed

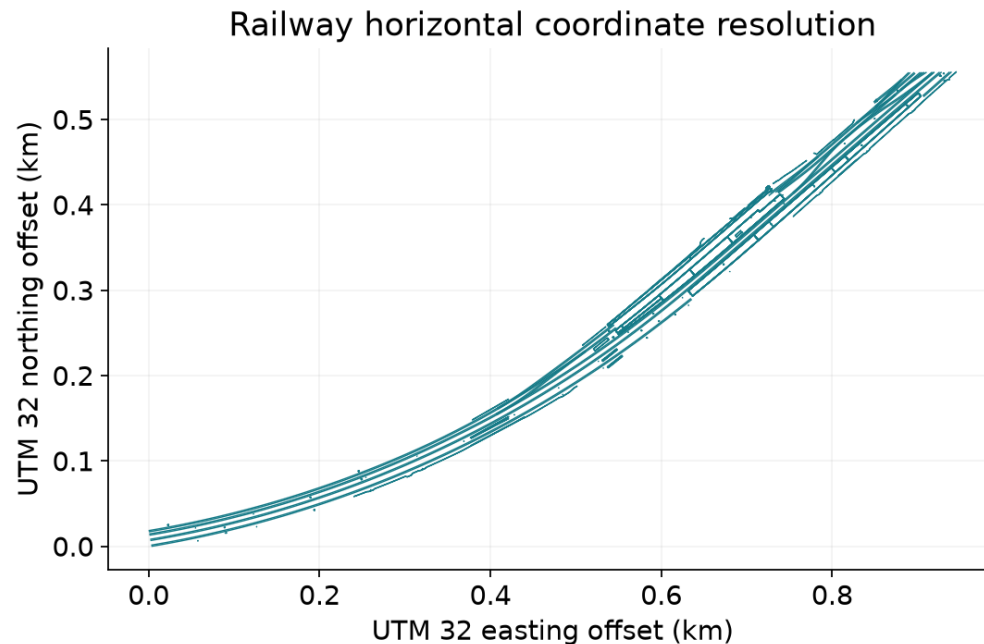
The original LandXML survey breaklines resolve and reach an ordinary USD bake rendered by Storm.

What it establishes

The case supports a site's calibrated CRS using partner survey lines. It does not establish independent survey accuracy.

Appendix: Useful horizontal placement without invented height

Can incomplete source metadata still support an honestly limited workflow?



Observed

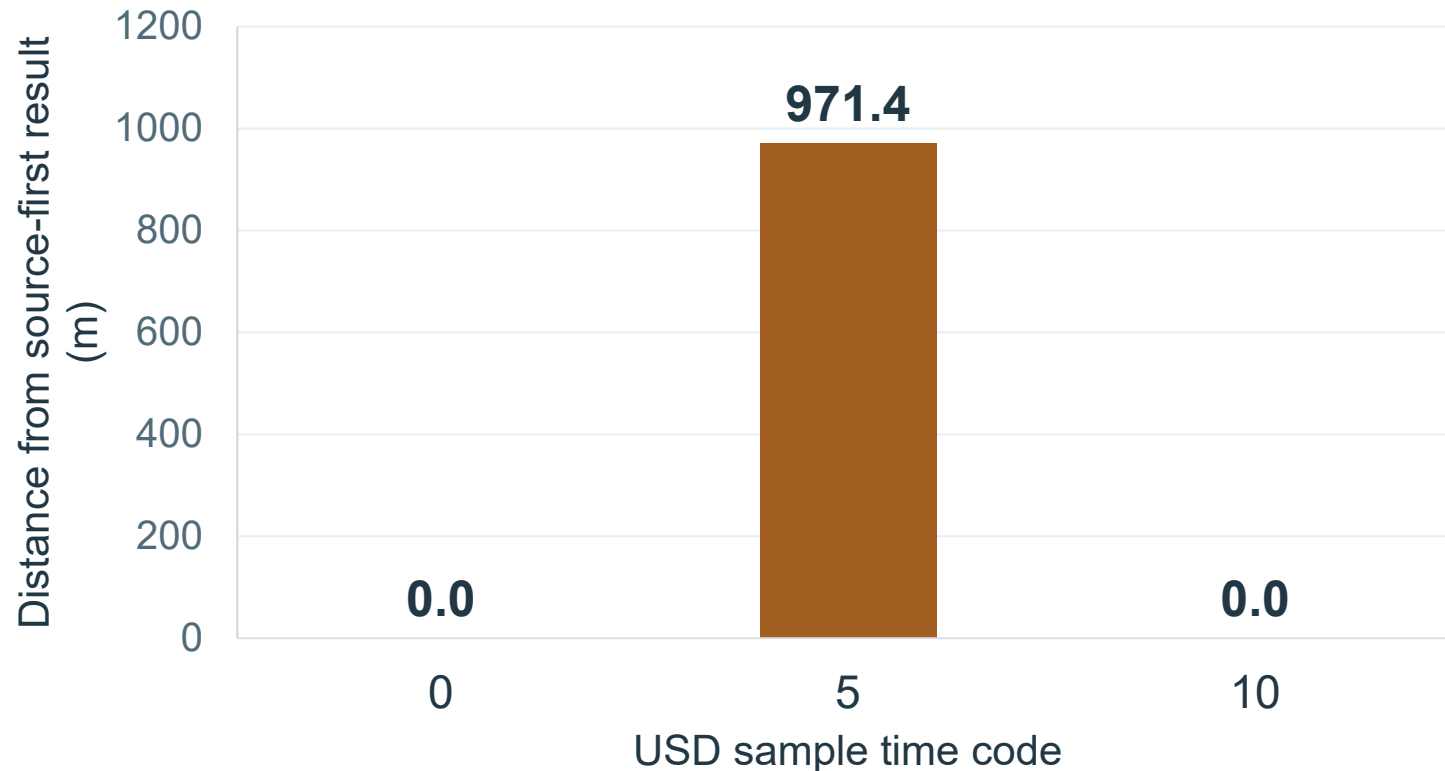
The horizontal railway copy resolves 15,822 vertices across 2,386 parts. The plot uses display-only offsets.

What it establishes

This case exercises visible limits on interpretation, not a silent assumption that every three-number tuple is a valid geographic 3D position.

Appendix: Animation evaluation order

**Interpolate the authored source location first, then convert coordinates.
Interpolating already-converted endpoints can produce a different path.**



Why the order matters

The distinguishing control makes the ordering rule testable. Preserving sample times alone cannot prove preservation of the original trajectory between samples.

Position interpolation is shown here. HPR wrap, held samples, reference offsets and value clips pass separate controls.

Appendix: Placement choices under review

Topic	Supported meaning and candidate detail
Physical orientation	Preferred HPR tuple in degrees, with explicit pose interpolation. Authored inputs only; adoption pending.
The object with its own georeference	Project adjustment meaning is supported. The detailed fixed working-frame convention remains a candidate.
Child objects	Model-local meaning is supported. The follow-up specifies chart/reset details and tests that offsets retain their interpretation.
Independently georeferenced repeated models	Keep a prototype's own geographic placement and apply each instance effect once.
Geographic queries and scene geometry	Return longitude/latitude/height for geographic queries. Cartesian geometry and bounds use the datum's Earth-centred frame.

Prefer HPR for authoring; review the precise pose convention and interpolation before adoption.

Appendix: Data and declaration choices

Topic	Supported meaning and candidate detail
Native data association	Name the asset, format, field and coordinate domain. Preserve the association between values, locations and observation times.
WKT string normalization	A prescribed text form supports token comparison. Different normalized texts can still describe equivalent CRSs.
Semantic CRS comparison	Specify equivalence criteria and the comparison domain, accounting for relevant axes, units and metadata.
Dependency declaration using Profiles	Declare the composed scene's need for geospatial interpretation, including unloaded content. Writers and assemblers maintain the declaration.
Baked output context	Record a Cartesian CRS and ordinary double-precision origin. Preserve precision and space/time coverage.

Coordinate epochs and scene-authored operation/resource controls remain deferred without foreclosing later support.

Appendix: Evidence and its limits

Path or check	Recorded evidence
Python and C++ readers	213,243 coordinate results per reader. Maximum measured difference 3.39e-08 m. They share PROJ and native dataset decoding.
Omniverse stage geometry integration	Reuses Python placement. Recorded readback covers 177,872 geometry vertices and 48 edit, failure and recovery checks, across 39 jobs.
Hydra / Storm	Two selected ordinary USD bakes reach native rendering. This consumer path uses the C++ placement results.
Recorded tests	88 regressions and 65 distinguishing controls passed. Fresh independent diagnostics verify six repairs and retain the unresolved normal case.
Source preservation	Original inputs and all frozen execution files remained unchanged during this run. The immutable receipt identifies the exact executed content.
Bake readbacks	11 selected exports were checked. The fresh nested-geometry counterexample verifies the complete inventory, including unexpected extras.

There are two placement readers. Omniverse reuses Python with a verified stage geometry sink. Hydra consumes a C++-resolved ordinary USD bake.